Math in Music Theory

Ellie Su

January 14, 2026

Roadmap (20 minutes)

1. Sound as a signal: amplitude vs. frequency
2. Pitch scale: octaves, semitones, and a logarithmic coordinate
3. Pitch classes: mod 12, transposition, and symmetry
4. Consonance: time re-alignment and harmonic overlap
5. Timbre: harmonics and Fourier series

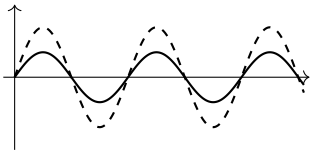
Sound as a Waveform

A simple model of a pure tone is a sine wave:

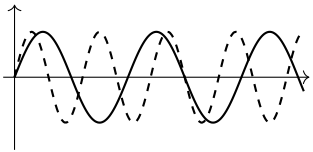
$$p(t) = A\sin(2\pi ft),$$

where:

- **p(t):** deviation of air pressure from equilibrium at a fixed point
- **Amplitude** A**:** size of vibration.
- **Frequency** f **(Hz):** cycles per second.
- **Period** T**:** time for one cycle ($T = 1/f$).



Same f**, larger** A louder, same pitch



Same A**, larger** f higher pitch

Pitch as a Logarithmic Coordinate

Doubling frequency raises pitch by one octave:

$$f \mapsto 2f \qquad (440\ \mathrm{Hz} \to 880\ \mathrm{Hz}).$$

In 12-TET, each semitone multiplies frequency by $2^{1/12}$.

Define *pitch height* by

$$\text{pitch height} = \log_2(f).$$

Consequences

$$f \to 2f \ \Rightarrow\ \log_2(f) \to \log_2(f) + 1 \quad \text{(octave)}$$

$$f \to 2^{1/12} f \ \Rightarrow\ \log_2(f) \to \log_2(f) + \tfrac{1}{12} \quad \text{(semitone)}$$

$$\log_2\left(\frac{f_2}{f_1}\right) = \log_2(f_2) - \log_2(f_1) \quad \text{(interval)}$$

Pitch Classes

In 12-TET, we treat notes an octave apart as the **same class** (same 'letter name'):

$$f \sim 2f \quad \Rightarrow \quad \text{collapse pitch to a loop.}$$

Label pitch classes by integers mod 12:

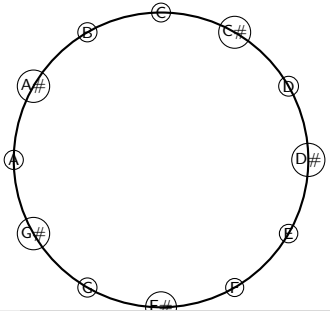
$$\mathrm{C} = 0,\ \mathrm{C\#} = 1,\ \ldots,\ \mathrm{B} = 11, \qquad \text{so } \mathbb{Z}_{12} = \{0, 1, \ldots, 11\}.$$

Musical meaning (transposition)

Moving up k semitones changes the class by

$$x \mapsto x + k \pmod{12}.$$

So 'wrap-around" is built in (B up 1 semitone $\to$ C).



Symmetries of the Pitch-Class Circle

Once pitches live on a 12-step circle, the natural moves are geometric.

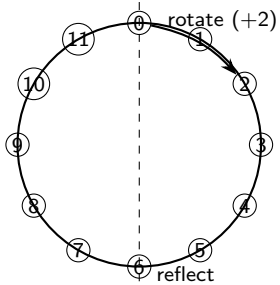
Two core operations

Transposition (rotation):

$$T_n(x) = x + n \pmod{12}.$$

Inversion (reflection):

$$I_a(x) = a - x \pmod{12}.$$



Musicwise:

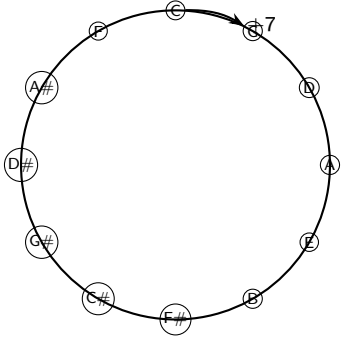
- rotation = shift every note up the same number of semitones n
- reflection = flip intervals around a center a

Circle of Fifths; One Special Rotation

A perfect fifth is $+7$ semitones:

$$x \mapsto x + 7 \pmod{12}.$$

Because $\gcd(7, 12) = 1$, repeating "$+7$" visits **all** pitch classes before returning.



Circle of Fifths

Intervals Are Ratios of Frequency

An **interval** between two pitches is the ratio

$$\frac{f_2}{f_1}.$$

Common stable-sounding intervals:

$$2:1 \text{ (octave)}, \quad 3:2 \text{ (perfect fifth)}, \quad 4:3 \text{ (perfect fourth)}, \quad 5:4 \text{ (major third)}.$$

Why ratios?

If you shift a melody up an octave (multiply all frequencies by 2), the ratios stay the same.

Audio example

Audio

Why Do Some Intervals Sound Consonant (Good)? (Reason 1: Time Re-alignment)

Suppose

$$\frac{f_2}{f_1} = \frac{p}{q} \quad \text{in lowest terms.}$$

- **Time re-alignment:** the combined waveform repeats exactly after a time T.

Smallest exact re-alignment time:

$$T = \frac{q}{f_1} = \frac{p}{f_2}.$$

Meaning: in time T, the lower completes q cycles and the higher completes p cycles.

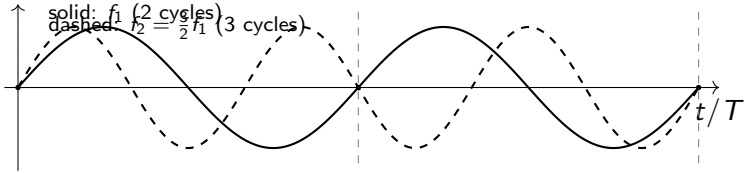
Heuristic

Small $p, q \Rightarrow$ short exact repetition $\Rightarrow$ simpler periodic structure (often less perceived "roughness").

Example: Perfect Fifth (3 : 2 Frequency Ratio) Re-aligns Quickly

$$\frac{f_2}{f_1} = \frac{3}{2}, \qquad T = \frac{2}{f_1} = \frac{3}{f_2}.$$

Over $[0, T]$: lower does 2 cycles; upper does 3 cycles.



Same idea works for 2:1, 4:3, 5:4 (small integers).

Preface: Harmonics

Real instrument notes are not perfect sines (like we modeled before). They include **harmonics** (overtones).

If the fundamental (main pitch) is f_0, harmonics occur at

$$f_0,\ 2f_0,\ 3f_0,\ 4f_0, \ldots$$

- f_0 sets the perceived pitch.
- Higher harmonics shape both **timbre (color)** and (often) perceived **consonance**.

Why Do Some Intervals Sound Consonant (Good)? (Reason 2: Overlapping Harmonics)

If $\frac{f_2}{f_1} = \frac{p}{q}$ (lowest terms), then

$$pf_1 = qf_2 \quad \text{and hence for any positive integer } k, \quad kp\, f_1 = kq\, f_2.$$

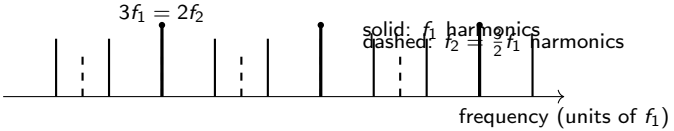
So a harmonic of the lower note can exactly match a harmonic of the higher note. Smaller order harmonics often have bigger impact on overall sound.

Example: Perfect Fifth (3 : 2 Frequency Ratio) Has Early Harmonic Overlap

If $\frac{f_2}{f_1} = \frac{3}{2}$ then

$$3f_1 = 2f_2,$$

so the **3rd harmonic** of the lower matches the **2nd harmonic** of the higher.



Why can the same pitch sound different depending on the instrument?

Timbre ("tone color") is why instruments can sound different on the *same pitch*.

Even if the fundamental pitch is the same (e.g. A4 at 440 Hz), instruments differ because they have different **harmonic mixtures**: different strengths of

$$f_0,\ 2f_0,\ 3f_0,\ \ldots$$

Demo (A4, 440 Hz**)**

1. Flute
2. Clarinet
3. Audio

What is Fourier?

Fourier idea: a periodic sound waveform can be written as a sum of sine waves.

For many musical tones, the main frequencies are the fundamental and its harmonics:

$$f_0,\ 2f_0,\ 3f_0, \ldots$$

Fourier series model:

$$p(t) = \sum_{n=1}^{\infty} A_n \sin(2\pi n f_0 t + \phi_n).$$

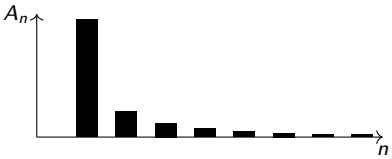
- f_0 sets the **pitch**.
- A_n (the amplitude of the nth harmonic) largely determines **timbre**.
- ϕ_n affects fine details of the waveform shape.

Accurate model of $p(t)$; we can describe it by "how much of each frequency is present": the **spectrum** $\{A_n\}$.

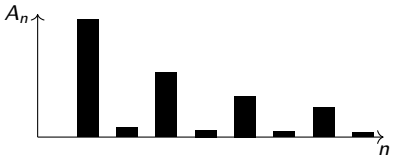
Example: Demo Flute v.s. Clarinet

Same pitch (f_0) but different harmonic strengths $\Rightarrow$ different instrument sound.

Mostly fundamental (flute-like)



Odd harmonics strong (clarinet-like)



Key message:

pitch is largely determined by f_0, timbre is largely determined by $\{A_n\}_{n\geq 1}$.

Summary

- Sound as a signal: a pure tone can be modeled by $p(t) = A\sin(2\pi ft)$; A controls loudness, f controls pitch, and $T = 1/f$.
- Pitch scale: octaves/semitones multiply f; using pitch height $\log_2(f)$ turns these multiplicative steps into additions (intervals become differences).
- Pitch classes on a 12-step loop: label notes by $\mathbb{Z}_{12}$; transposition is $x \mapsto x + k \pmod{12}$, and inversion is $I_a(x) = a - x \pmod{12}$.
- Circle of fifths: the $+7$ rotation $x \mapsto x + 7 \pmod{12}$ cycles through all pitch classes since $\gcd(7, 12) = 1$.
- Consonance (thread: 3:2): rational ratios re-align in time ($T = \frac{q}{f_1} = \frac{p}{f_2}$) and create early harmonic overlap (e.g. $3f_1 = 2f_2$).
- Timbre: instruments differ by harmonic mixtures; Fourier series models a tone as $p(t) = \sum_{n\geq 1} A_n \sin(2\pi nf_0 t + \phi_n)$, where $\{A_n\}$ (the spectrum) largely shapes timbre.